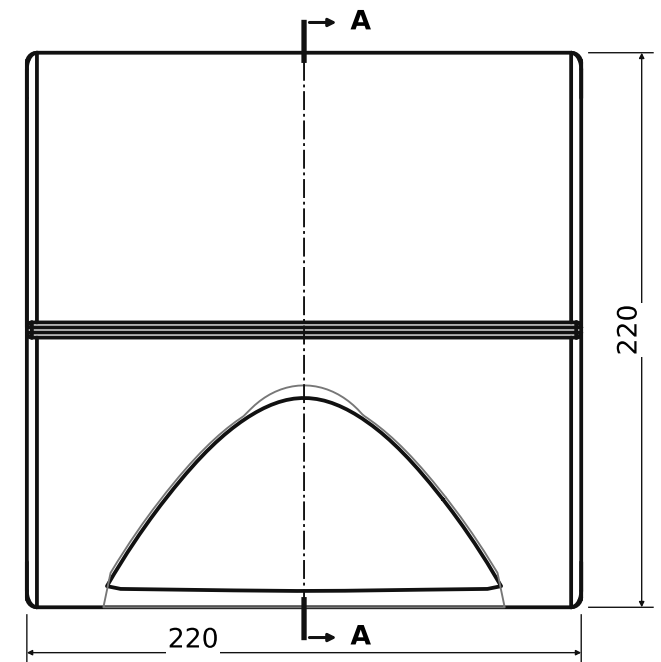
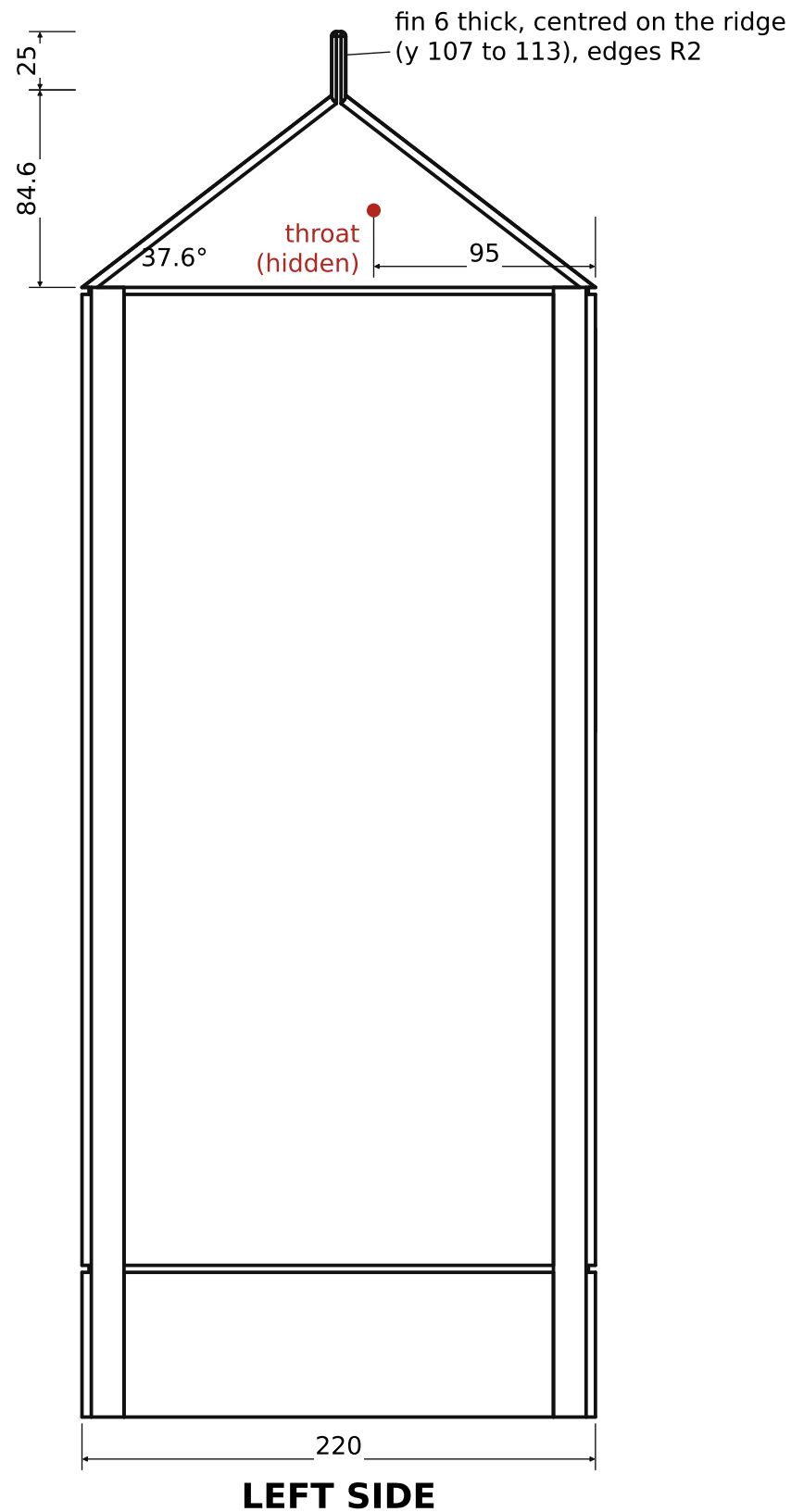
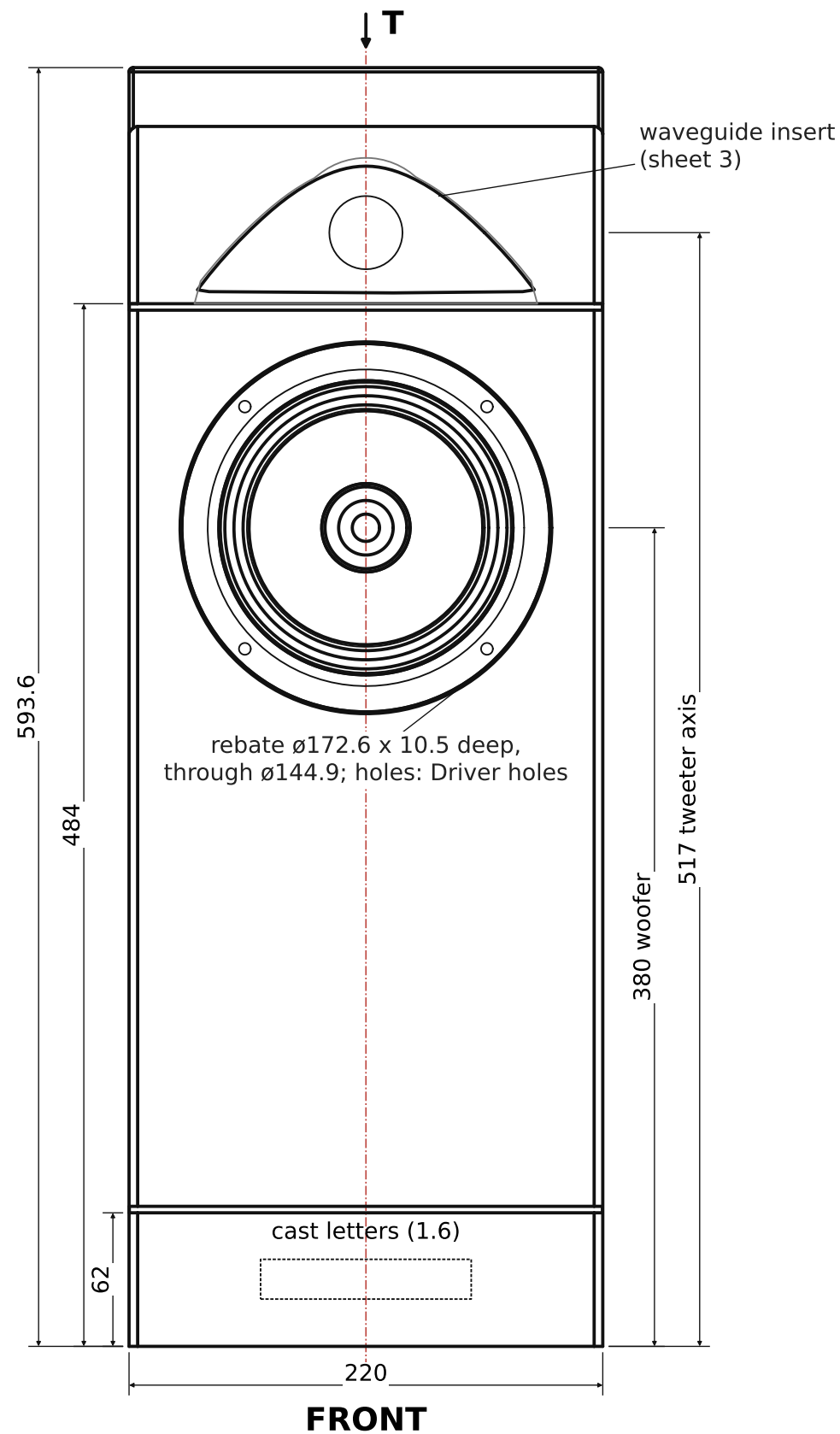




VIEW T (FROM ABOVE)

black: the waveguide's mouth, where its lip meets the roof
grey: the insert's seam

Notes: notes 1.1 to 1.6, sheet 7

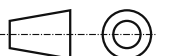
Driver holes (front baffle, from its front face)

Woofer, centre z 380 on the centreline (M2)

rebate $\varnothing 172.6 \times 10.5$ deep; through $\varnothing 144.9$; 4 x $\varnothing 5.5$ on $\varnothing 159$, the first 45° above the horizontal, 90° apart, for M4 T-nuts (1.2).

General arrangement

earmilk bookshelf. Front, left side and top at 1:3. Heights from the floor; depths from the front face.

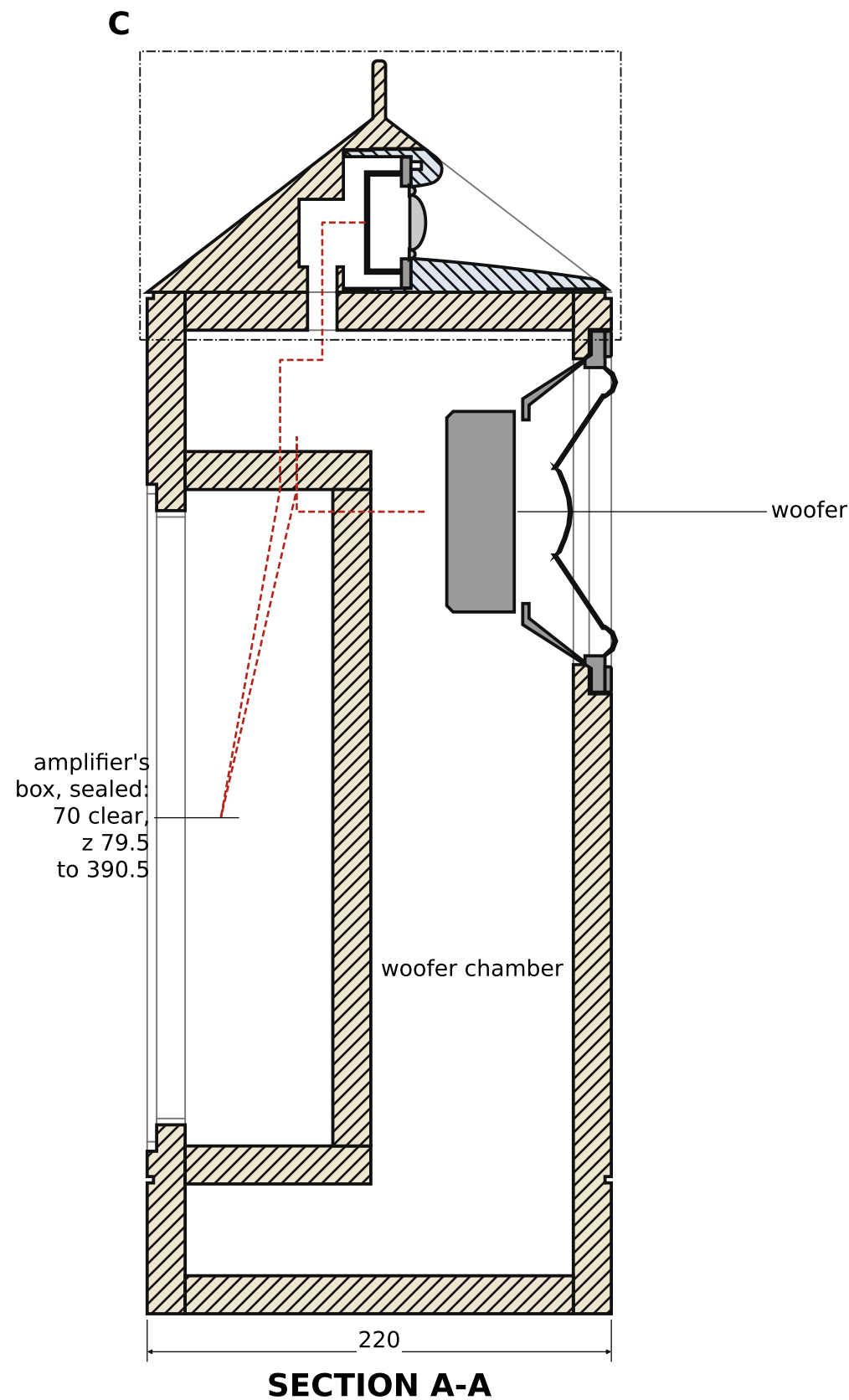


mm | issue 2026-10-09 | HOLD until M2 (the drivers)

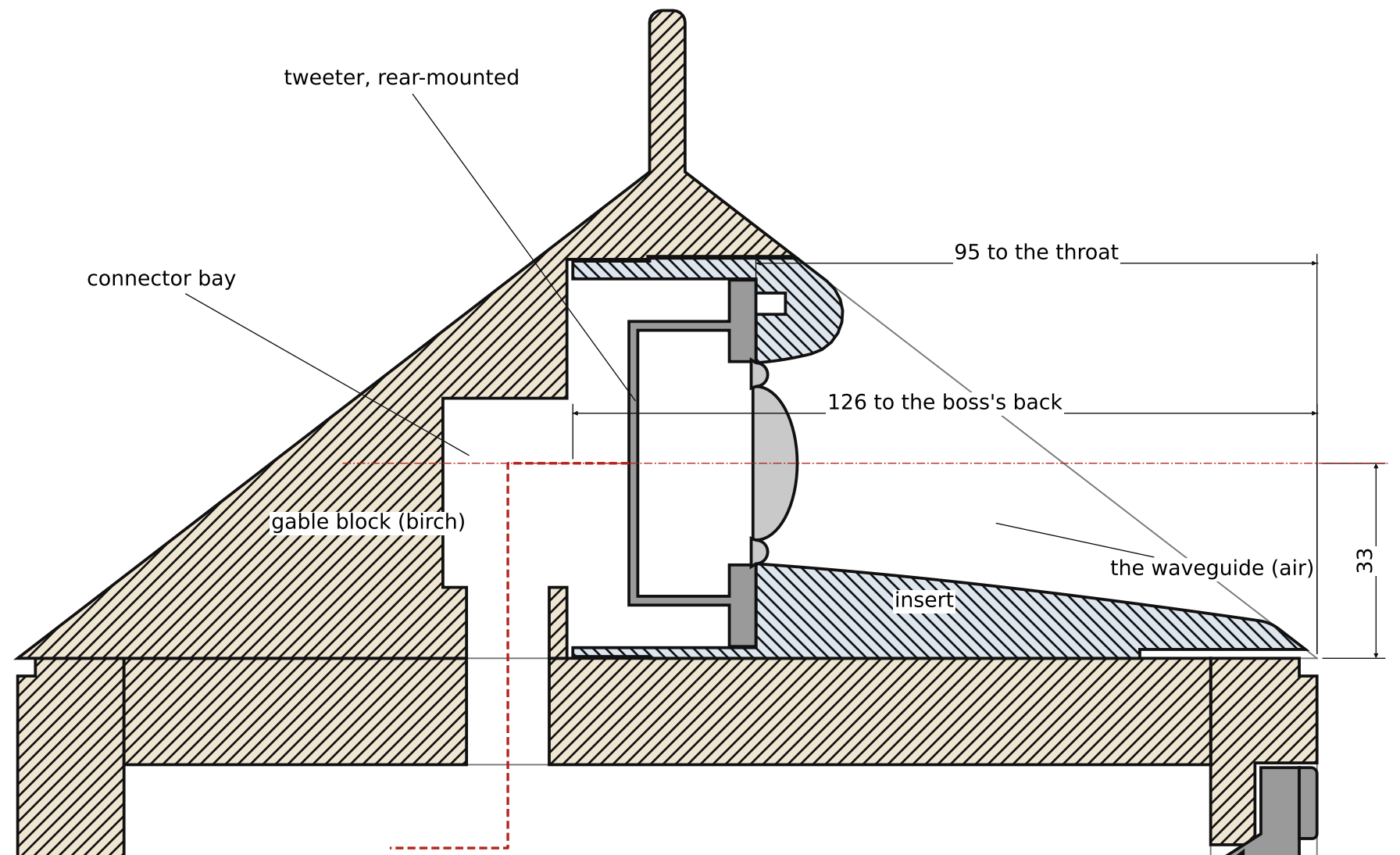
Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

1 of 9

EM-BS-001 rev D



red dashed: the cables' runs to the amplifier (sheet 4)

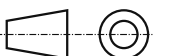


the cable simplified: its connector and the tweeter's own lead on sheet 3

The roof: notes 2.1 to 2.4, sheet 8

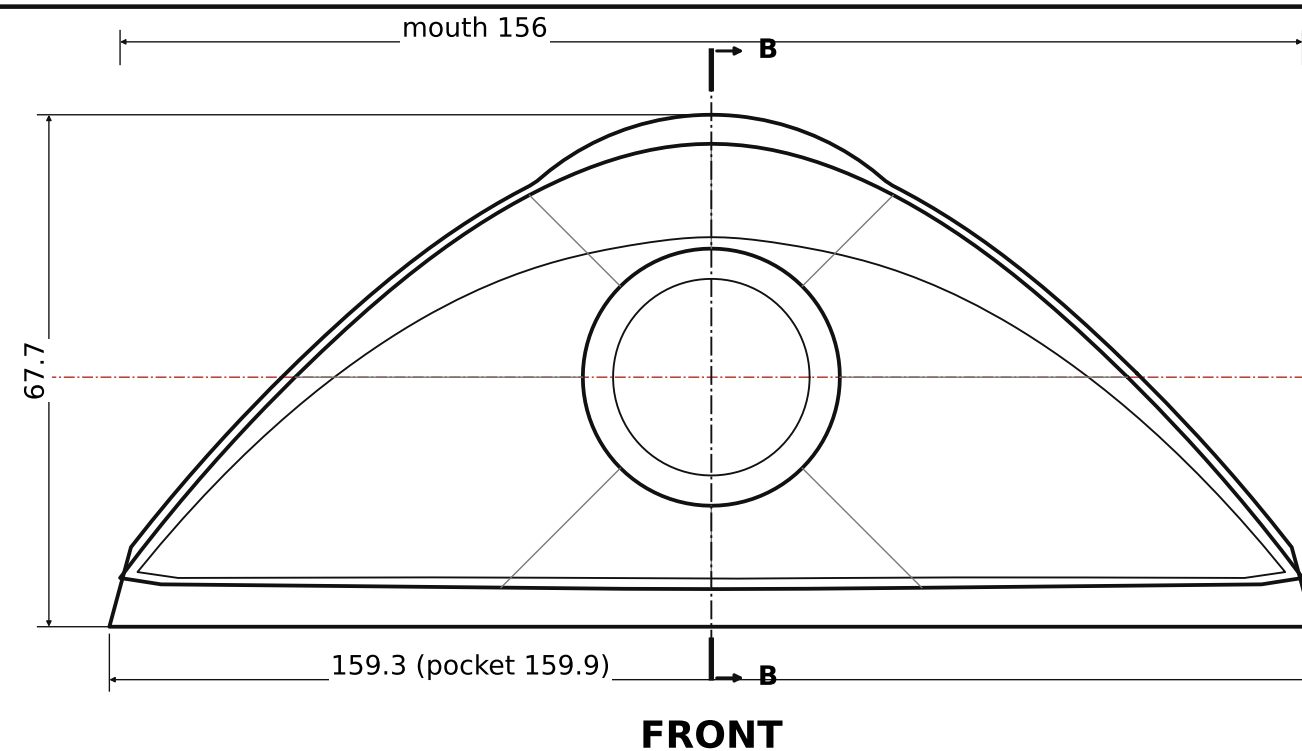
Section on the centreline

earmilk bookshelf. Section A-A on the centre plane (x 110), from the left, at 1:3; the roof at 1:1. The insert hatched the other way.

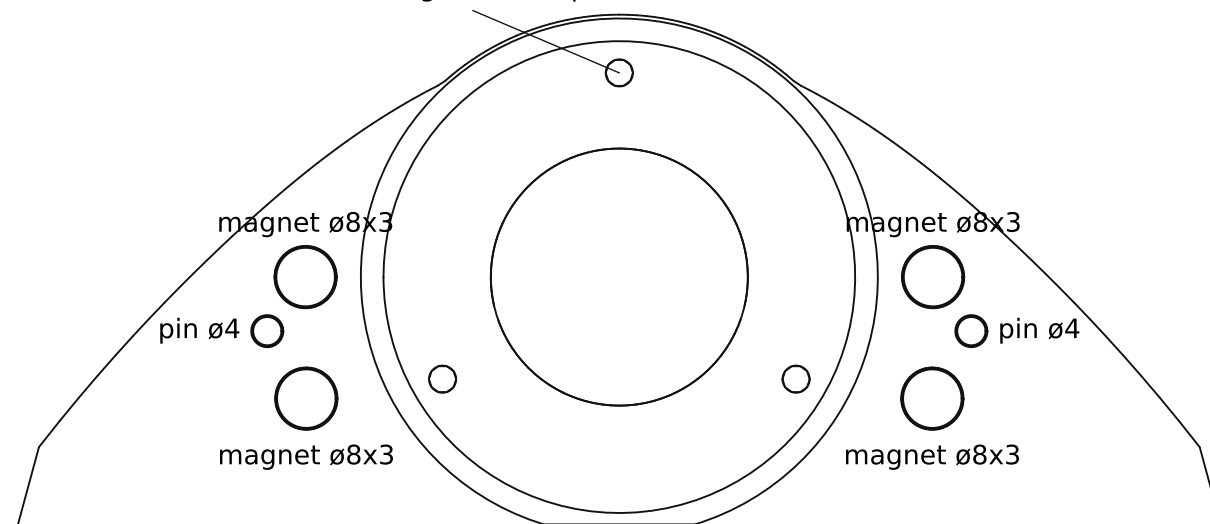


mm | issue 2026-10-09 | for reference
Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

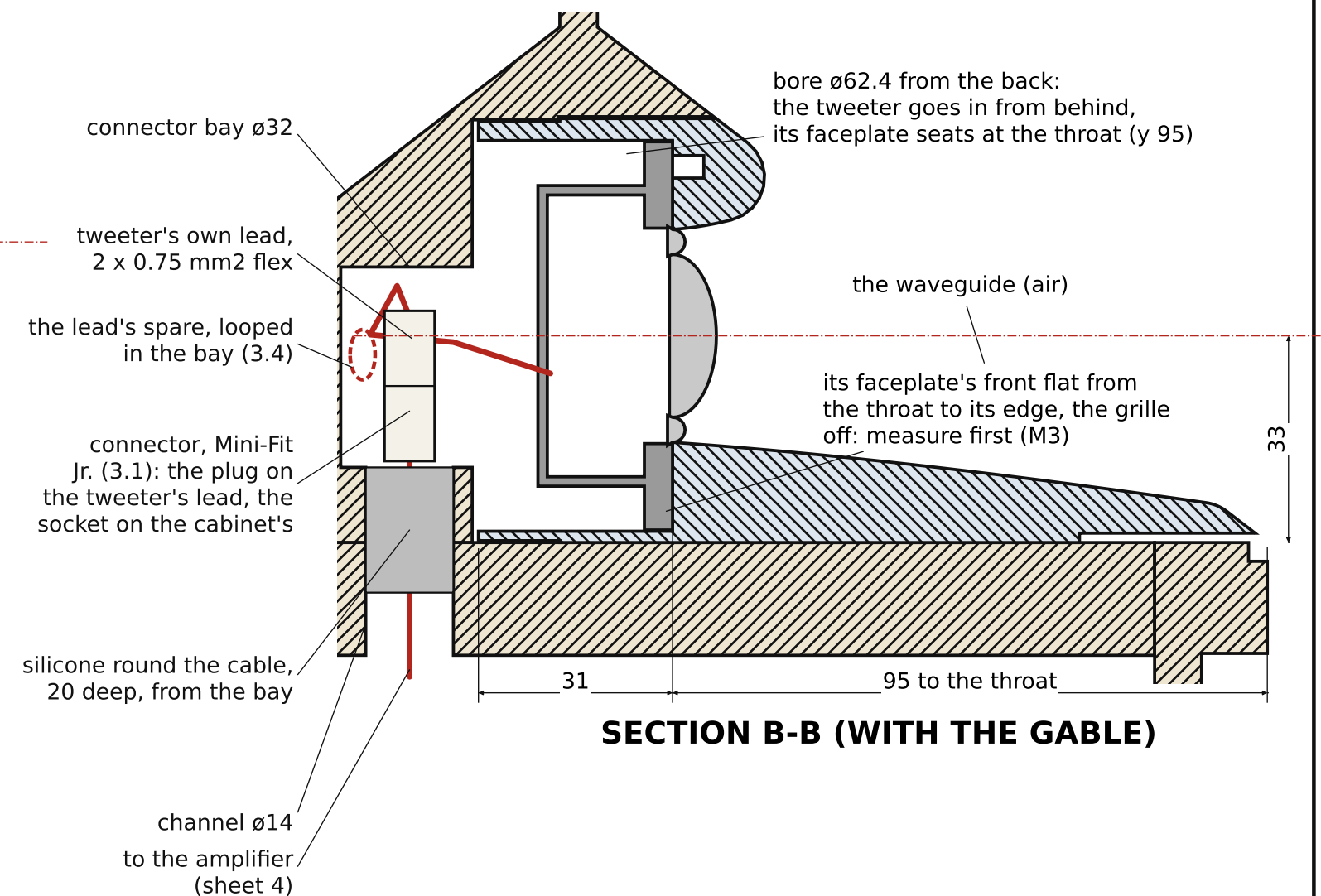
2 of 9
EM-BS-002 rev D



3 x $\varnothing 3.6 \times 5$ holes on $\varnothing 54$ in the seat, one at the top, 120° apart, for M2.5 x 4 knurled brass inserts bonded with epoxy; M2.5 x 8 button heads (ISO 7380, A2) through the faceplate: measure it first

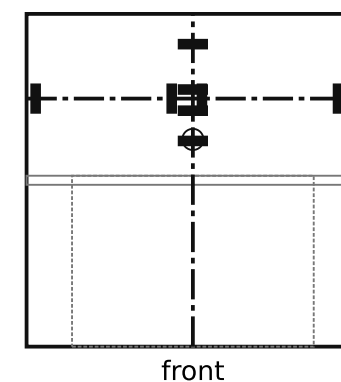


BACK (THE FACE THAT MEETS THE POCKET)



Fitting the tweeter and the insert: notes 3.1 to 3.9, sheet 9

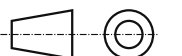
PRINTED GABLE (ROUTE B), PLAN, 1:5



Split planes (chain lines) and pins (bars): stl/gable-print/
 x = 110: (y 136, z 540), (y 156, z 490), (y 170, z 510), (y 200, z 490)
 y = 164: (x 6, z 490), (x 96, z 510), (x 116, z 490), (x 206, z 510)
 8 steel pins $\varnothing 4 \times 20$ a speaker, in $\varnothing 4.2$ holes 10.5 deep each side; dry-fit each joint closed, then 3M DP420 over its full rim, and seal the channel halves' seam along its length. Hollow the pieces to 3 mm walls, two $\varnothing 3$ drains each in the outer slopes or end faces only (none in the base, a joint face, the channel, the bay or the pocket), plugged with resin and filled before paint.

Waveguide insert and tweeter mount

earmilk bookshelf. The insert at 1:1: front, section on the axis, back; the pocket; fitting and service.

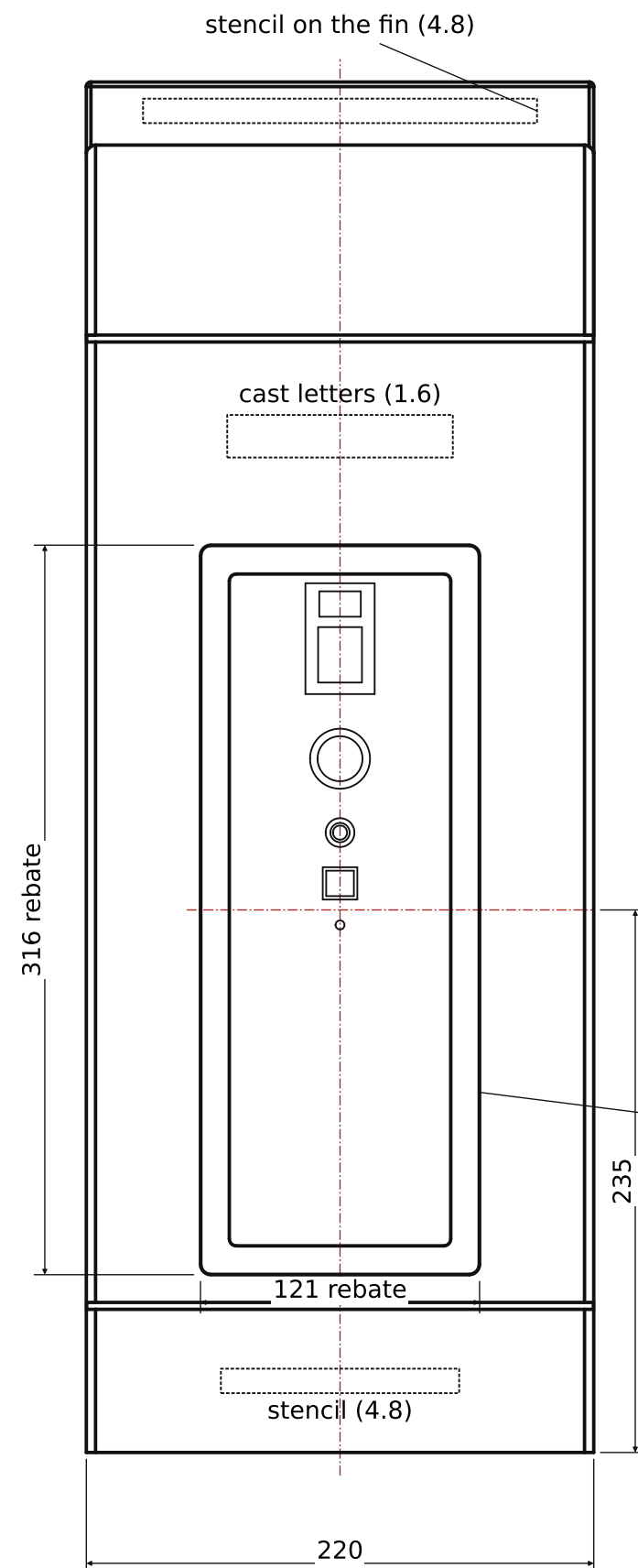


mm | issue 2026-10-09 | HOLD until M3, M5 (tweeter, connector)

Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

3 of 9

EM-BS-003 rev D



BACK

The Nutrition Facts go on the right side: 150 x 147, top at z 420, centred across its depth, 35 from each vertical corner.

Mains
IEC C14 inlet, switch

Signal in
XLR or RCA
USB for the DSP

Hypex FusionAmp FA122
2 channels, DSP crossover and EQ
125 + 125 W into 4 ohm
upright on the back
in its own sealed box (not airtight itself)
filters: Hypex Filter Design, over USB
mains in on the plate (IEC C14 with switch)

CH1
CH2

Tweeter, Scan-Speak D3004/602200
26 mm dome, faceplate behind the insert
1 mm2 pair, red +, black -

Woofers, SB Acoustics SB17NRX2C35-8
6.5 in, sealed, DSP shelf to 45 Hz
1.5 mm2 pair, red +, black -

Runs, connectors and seals: notes 4.1 to 4.8, sheet 8

plate 120 x 315, in a
121 x 316 x 4.5 rebate;
cut-out 96 x 291, R3 (4.1)

Back and wiring

earmilk bookshelf. The back at 1:3 and the amplifier, drivers, cables and connectors.
Active: the amplifier's DSP is the crossover.

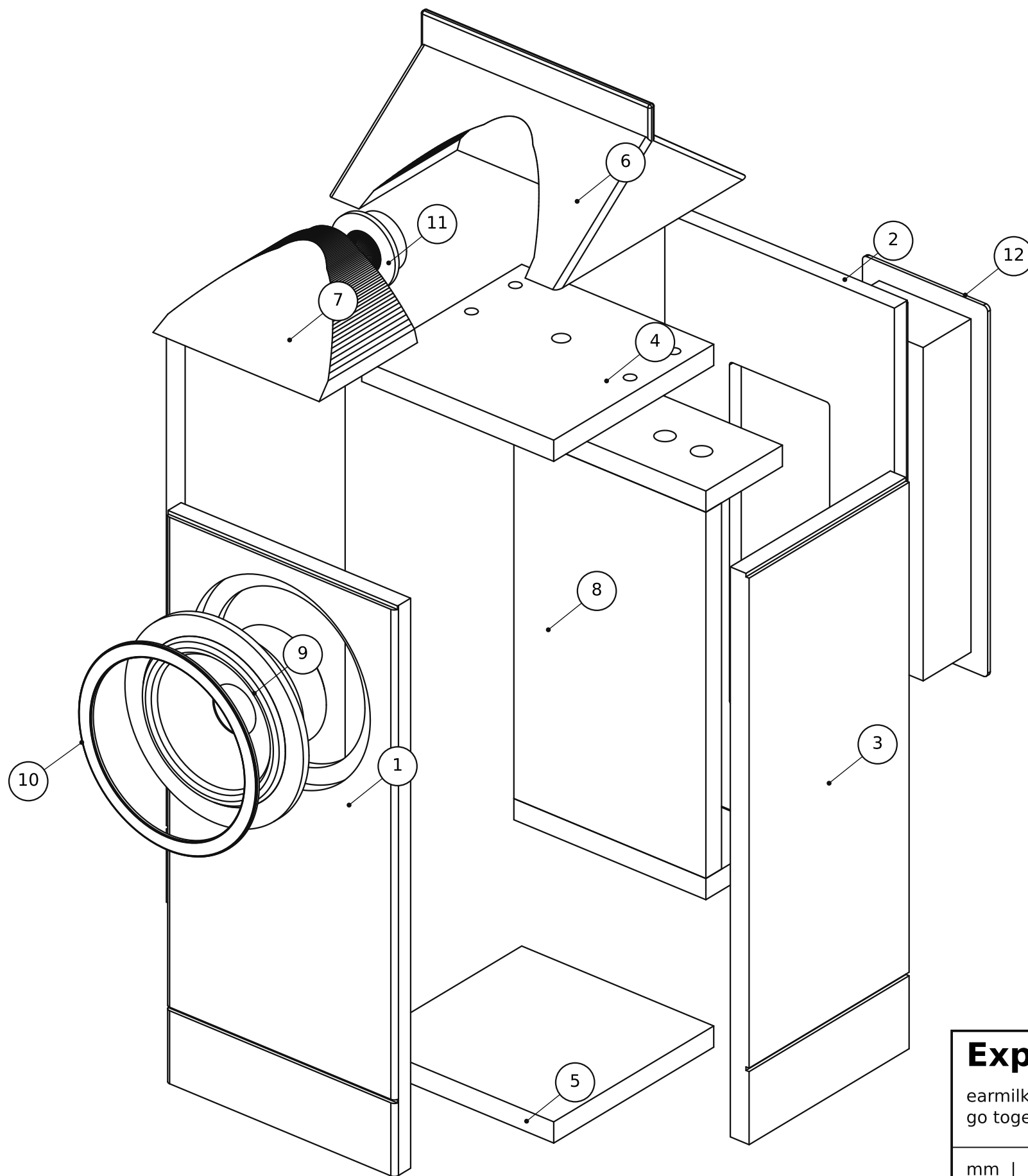


mm | issue 2026-10-09 | HOLD until M4 (the amplifier)

Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

4 of 9

EM-BS-004 rev D



Parts

- ① **Front baffle x1**
18 mm Baltic birch, CNC: the woofer's cut-out, rebate
- ② **Back x1**
18 mm birch: the amplifier's cut-out and rebate
- ③ **Sides x2**
18 mm birch; the Facts printed on the right one
- ④ **Top panel x1**
18 mm birch: the 14 mm cable hole, 4 dowel holes
- ⑤ **Bottom panel x1**
18 mm birch
- ⑥ **Gable block x1**
laminated birch, CNC, or printed in four pieces (step 1)
- ⑦ **Waveguide insert x1**
SLA tough resin or MJF nylon; magnets, pins
- ⑧ **Amplifier box x3**
birch floor, lid (2 glands) and front; sealed
- ⑨ **Woofer x1**
SB Acoustics SB17NRX2C35-8, 6.5 in
- ⑩ **Trim ring x1**
printed, satin black, over the frame and screws
- ⑪ **Tweeter x1**
Scan-Speak Illuminator D3004/602200
- ⑫ **Amplifier x1**
Hypex FusionAmp FA122, DSP, 2 channels

Glue-up order: G1 to G4, sheet 8

Fixings, a speaker (bom.csv)

M4 T-nuts and M4 x 20 low button heads (F1)	4
M16 x 1.5 cable glands and their lock nuts	2
magnets ø8 x 3 (insert and pocket)	8
pins ø4 x 8	2
fluted dowels 10 x 28 (F3)	4
Molex Mini-Fit Jr. 2-circuit plug and socket (3.1)	1 pair
M2.5 x 4 knurled brass inserts and M2.5 x 8 button heads	3
steel pins: ø4 x 20 (printed gable)	8
foam gaskets 1.5 and 0.5 mm, EPDM tape 3 mm (the plate)	a strip each
4.3 x 16 self-tapping pan heads (the plate)	as drilled

Exploded view and parts

earmilk bookshelf. Isometric, not to scale: the parts pulled apart along the way they go together, and what each one is.

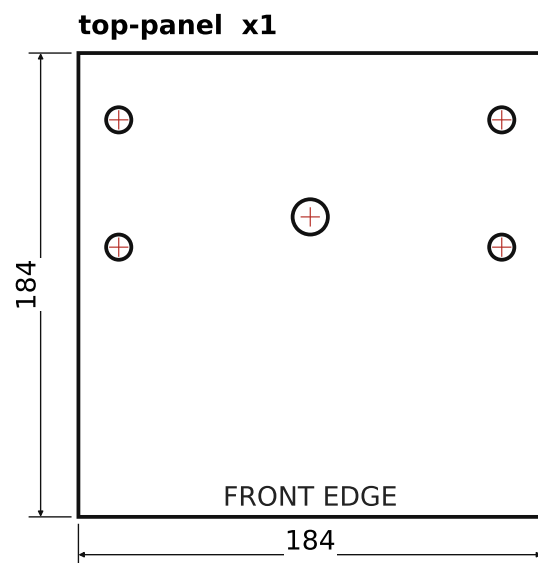


mm | issue 2026-10-09 | for reference

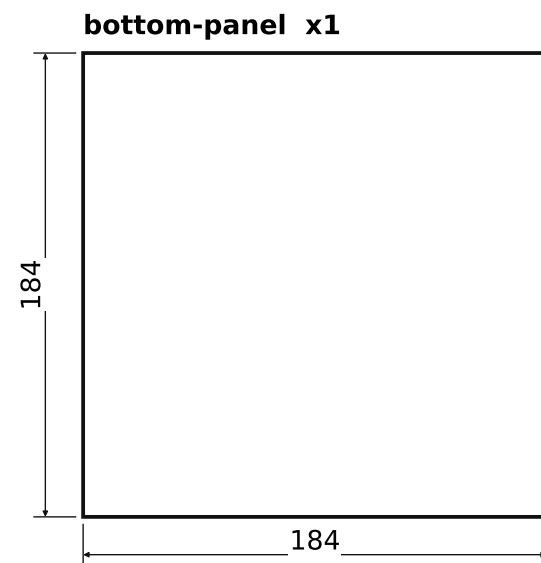
Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

5 of 9

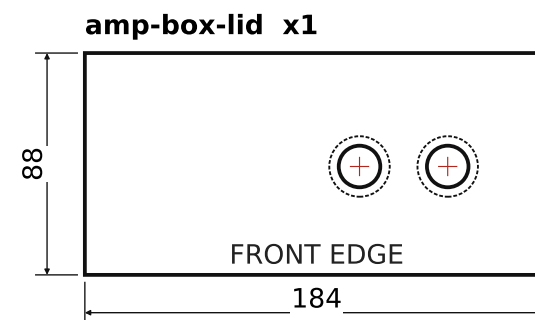
EM-BS-005 rev D



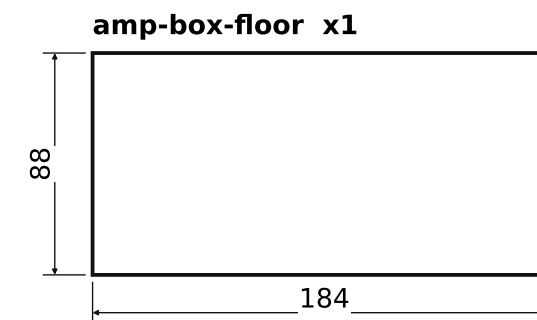
ø14 at (92, 119): cut inside
4 x ø10 at (16, 107), (168, 107), (16, 157.5), (168, 157.5): drill ø10, 10 deep
upper face up, FRONT EDGE to the open front (the ø14 hole 119 from it and 65 from the back): turned round, the channel opens under the insert and the dowels miss the gable; dowels register the gable block; the tweeter cable's hole is sealed with silicone from the bay after wiring



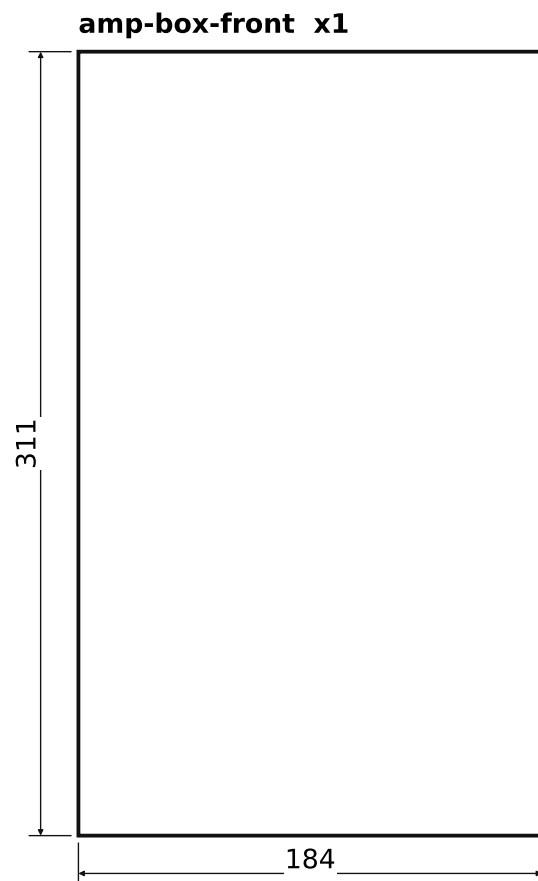
either face



2 x ø16.5 at (144, 43), (109, 43): cut inside
2 x ø24 x 8 from the underside (dashed): amp-box-lid-underside.dxf, the panel turned over left to right, at (40, 43), (75, 43)
the amplifier box's lid, z 390.5 to 408.5, its FRONT EDGE over the box's front; the glands and their nuts: note 4.6

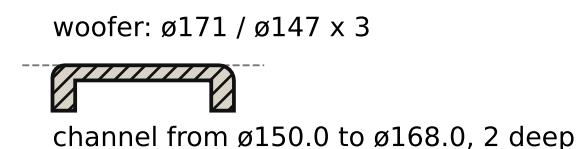


the amplifier box's floor, z 61.5 to 79.5, against the back



the amplifier box's front, between floor and lid, 70 in front of the back's inner face; glue and seal all round

Trim rings, section at 2:1 (stl/trim-ring-*.stl): flat, flush with the finish



the channel clears the M4 x 20 heads (F1): 8 across, 1.6 high or less

Panel details

earmilk bookshelf. At 1:3: each panel seen from its outer (upper) face, from its lower-left corner as its DXF; holes from that corner. 18 mm birch.



mm | issue 2026-10-09 | cut after M1; the trim rings HOLD until M2
Tolerances: panels ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

6 of 9
EM-BS-006 rev D

Measure first

- M1

Ply: 18 thick; the sides and inner panels are 220 - 2 x 18 = 184 wide. For a real thickness t: 220 - 2t (heights, pockets and rebates unchanged); set WALL = t in params.py and run fab/build.sh again. The gable's layers are t too: 7 for its 109.6 (one more under t 15.66). The birch under each rebate is t less its depth (woofer t - 10.5): buy T-nuts whose barrels are 0.5 shorter than that (a CAD check).
- M2

Woofer (SB17NRX2C35-8): flange 6.5 thick (SB's drawing, its labels not seen: measure the flange at its screw holes and anything standing above it; over 6.5 at the rim, stop: a flush rebate would leave 3.1 of birch, too little for any T-nut) sets the rebate, 10.5 deep; the frame's diameter + 1.6 sets the rebate's ø172.6; the cut-out ø144.9; 4 holes on ø159; the surround at its glue line + 2 sets the trim ring's bore, ø147.
- M3

Scan-Speak Illuminator D3004/602200: the faceplate ø62 sets the bore, ø62.4; its hole circle, ø54, and its body, ø48.3 or less, set the seat's screws; its thickness t (4.5) sets their length, M2.5 x (t + 3.5); its front must be flat from ø34 to ø62 and the grille off (or a seat recess for it); the dome and surround 34 across or less. The seat's holes take M2.5 x 4 knurled brass inserts 3.5 or less across, bonded: each hole is the insert's outside diameter + 0.1 (ø3.6 for 3.5). Shops give its cut-out 47.8 to 48, three ø3.3 holes, 45.3 deep (from where?). Go or no-go: its body 26 or less behind the faceplate's rear face (the bore runs y 99.5 to 126); over that, print nothing: the owner chooses the fallback (README). Its solder tabs: measure each one's angle from the nearest hole and its radius from the axis; each head (ø4.7 on ø54) needs 1.0 clear of a tab and its solder, or bond the faceplate to the seat with epoxy instead of screwing it. Print nothing until it is measured.
- M4

Hypex FusionAmp FA122 (Hypex's manual: vertical or horizontal, in its own compartment; here upright): the plate 120 x 315 sets the rebate, 121 x 316; its corner radius r sets the rebate's corners, R = r + 0.5 (R4.5 for r 4: a square corner, or one under R2.8, will not seat); its thickness t (3) sets the rebate's depth, t + 1.5 (4.5, over the EPDM pressed), and the screws' length, t + 13 (16); the module sets the cut-out, 96 x 291; the screw holes come from the plate in hand.
- M5

Molex Mini-Fit Jr., 2 circuits pair (39-01-2020 and 39-01-2021): its mated length L (24 drawn, from no maker's drawing: Molex SD-5557-003 and 55590020-SD give it) sets the bay, ø32, which leaves 8 over the mated pair for the lead's bend. Print the test bay first (stl/connector-test-bay.stl, README, step 1) and mate and unlatch the pair in it by hand.
- Cut

Cut once M1 is entered: the sides, the bottom and top panels and the amplifier box's panels (sheet 6). On hold until measured (HOLD in their title blocks and DXFs, stl/HOLD.txt): the front (M2), the back (M4), and every print: the insert, the gable, and the trim rings.

Sheet 1: Notes

- 1.1

Cabinet: 18 mm Baltic birch, glued. The vertical corners and the gable's hips are rounded 4 mm, the fin's edges 2 mm. 3 x 3 mm shadow lines at z 62 and z 481.
- 1.2

The drivers sit flush: each rebate is the measured flange + 1.0 (a 1.5 mm closed-cell foam gasket, pressed) + 3 (the ring): woofer 10.5 for a 6.5 flange; measure each flange before cutting (M2): the rebate is the flange + 4.0 only while the T-nut's barrel + 0.5 of birch stays under it: 7.5 under the woofer's (barrel 7 or less). The screws and T-nuts as F1, pressed in before the glue-up. A printed trim ring over each flange (stl/trim-ring-*.stl, woofer ø171 / ø147 x 3, flat, a channel in its back over the screws' heads: the sections at 2:1 on sheet 6) held by three 5 mm dots of neutral-cure silicone, 0.8 reveal all round. Its inner ø is the surround at its glue line + 2: measure the drivers and print the rings last (the woofer's ring, ø171, needs a bed of 180 or more).
- 1.3

The tweeter is at the throat of a waveguide insert in the roof (sheet 3): axis z 517, 95 behind the front face. The shape is set by simulation (fab/out-bookshelf/acoustics/waveguide).
- 1.4

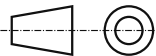
The back (the amplifier) is drawn on sheet 4 with the wiring; the Nutrition Facts are printed on the right side.
- 1.5

Finish: colour coat and 2K clear; the Facts printed between them.
- 1.6

The cast letters (out-bookshelf/metal/wordmark-letters.*), 1.2 thick and proud: 97.8 x 18.4 of ink, centred across the plinth's front, the badge's centre z 31; on the back the same at z 440. Place and drill them from the 1:1 templates, out-bookshelf/metal/wordmark-template-front.pdf and -back.pdf, after the clear coat.

Notes: measure first; sheet 1

earmilk bookshelf. What to measure before cutting or printing, and the notes sheet 1 point to by number.



Sheet 2: The roof

- 2.1 The waveguide insert (light blue) sits in a pocket in the gable block, 0.3 mm clear all round: the pocket +0.2/0 and the insert 0/-0.15 on their widths, so 0.30 to 0.48 a side. Paint only the insert's face: mask its sides, base and back and the pocket's walls. It slides out forward, level, with the tweeter on it.
- 2.2 The tweeter is rear-mounted: it goes in from behind through a $\varnothing 62.4$ bore and its faceplate seats on the ring round the throat, its dome and surround filling the 34 mm throat, screwed from behind through its own holes (sheet 3).
- 2.3 Its lead leaves the boss's open back into the connector bay behind it ($\varnothing 32$ x 22 deep), where it plugs into the cabinet's lead; that runs down a 14 mm channel through the block and the top panel, sealed round the cable with neutral-cure silicone from the bay.
- 2.4 Four magnets and two pins in the insert's back face meet their partners in the pocket's back wall (sheet 3).

Sheet 4: Runs, connectors and seals

- 4.1 The amplifier's plate (120 x 315 x 3) lies on 3 mm EPDM tape in its 4.5 deep rebate, R4.5 corners, its module through the 96 x 291 cut-out (R3 corners: a 6 mm cutter or smaller). Drill its screw holes $\varnothing 3.5$ through the 13.5 left under the rebate, from the plate in hand, centred in its flange, 8 or 10 to suit it: 4.3 x 16 self-tapping pan heads (Hypex's 4.3 x 25 kit also works, its tips 7 mm into the sealed box).
- 4.2 Woofer (CH1): from its gland in the amplifier box's lid to the woofer's terminals, 0.4 m of 1.5 mm2.
- 4.3 Tweeter (CH2): from its gland up behind the woofer's magnet to the top panel, through the 14 mm channel in the top panel and the gable block, 0.6 m of 1.0 mm2, ending 40 above the top panel's underside in the socket of the connector (3.1, sheet 3) standing on the bay's floor, fed down from the bay (3.4).
- 4.4 Seal the tweeter cable in its channel with 20 mm of neutral-cure silicone from the bay (sheet 3): the box is sealed, and the insert's pocket is open to the room through its seam.
- 4.5 Round-sheathed cable (H05VV-F 2 x 1.5 and 2 x 1.0 mm2, about 8.0 and 6.4 across), so each gland grips its cable and the silicone seals round the tweeter's. Inside the amplifier's box cut each of Hypex's harness pairs to about 150 mm and butt-splice it to its round cable (woofer 1.5, tweeter 1.0 mm2; crimped splices under heat-shrink): only round cable passes the glands. Tie the woofer's run to the amplifier box's lid beside its gland and to the box's front face at z 380 (adhesive cable-tie mounts); the tweeter's run is about 60 mm long and needs no tie.

- 4.6 The amplifier box's lid: 2 M16 x 1.5 long-thread (15 mm) cable glands, one per cable (clamping about 4.5 to 10), in $\varnothing 16.5$ holes 144, 109 from the left side's inner face and 43 from the lid's front edge. Their lock nuts, 20 AF or less (23 across corners) and 5 thick, are epoxied into the $\varnothing 24$ x 8 counterbores under the lid before it goes in (G2), each centred on a gland's body screwed through it and taken out again: a spanner cannot reach a nut sunk 3 deep. After the finish screw the bodies into them from above, feed the cables, then tighten the domes.
- 4.7 The crossover (2.4 kHz, LR4), the woofer's shelf to 45 Hz (+8.3 dB) with a 35 Hz high-pass under it, a limiter and the delays are set in the DSP (out-bookshelf/dsp/). Wire red to + throughout.
- 4.8 The stencils (out-bookshelf/marks/stencil-*.svg), sprayed through vinyl masks before the clear: SHAKE WELL, 15 mm type, centred on the plinth's back at z 31; OPEN OTHER SIDE, 15 mm type with its arrow, centred on the fin's back face.

Glue-up, in order

- G1 Cut the bottom panel and every inner panel at one fence setting, 184 ± 0.2 wide (tighter than the title block's ± 0.5 : they set and fill the sides' spacing). Lay the back face down. Glue both sides and the bottom to it; clamp across the sides and check the diagonals.
- G2 The amplifier box: epoxy the glands' lock nuts into the lid's counterbores first (sheet 4); then its floor, front and lid between the sides, against the back, the joints sealed; clamp across the sides.
- G3 The top panel, dowel holes up and its FRONT EDGE to the open front (sheet 6), no cable-tie mounts on its underside, slid in from the front and glued to the sides and the back; clamp across the sides.
- G4 Plane any inner panel's front edge that stands proud of the sides flush. Press the T-nuts into the front baffle's inside face; glue the baffle on last, onto the sides, the bottom, the top panel and every inner panel; clamp front to back. When the glue has set, sand the front's top edge flush with the top panel (F2), then fix the gable block (F3).

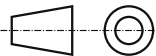
Fixings and the pocket's floor

- F1 Drivers: M4 x 20 low-profile (wafer) button head, A2, the head 1.6 high or less and 8 across or less (ISO 7380 is 2.2 high: the trim ring's channel is 2 deep). M4 T-nuts, flange 12 or less across (woofer) (DIN 1624's M4 is 15: grind each flange flat 6.0 from its centre on the side facing the cut-out), barrel 7 or less (woofer), pressed in before the glue-up.

- F2 Before the gable block goes on, sand the front panel's top edge flush with the top panel to 0.1: the insert sits on both. Mask the pocket's floor with the insert's sides, base and back when you spray: paint only the insert's face.
- F3 The gable block on four 10 x 28 fluted dowels (10 deep in the top panel, 20 in the block): a birch block with wood glue; a printed one with epoxy (3M DP420) on its base, scuffed with 120 grit, and on the dowels: wood glue does not hold cured resin.

Notes: sheets 2 and 4 to 6; glue-up

earmilk bookshelf. The notes sheets 2 and 4 to 6 point to by number; the order of the glue-up; the fixings.



mm | issue 2026-10-09 | notes

Tolerances: panels ± 0.5 , pockets ± 0.2 , holes +0.2/0, prints ± 0.15

8 of 9

EM-BS-008 rev D

Sheet 3: Fitting the tweeter and the insert

- 3.1 Solder the tweeter's own lead (220 mm of 2 x 0.75 mm2 silicone-insulated flex, twisted (about 2.2 mm each)) to its tabs and crimp the connector's plug on its end: Molex Mini-Fit Jr., 2 circuits, plug housing 39-01-2021 with male terminals 39-00-0041 (18 to 24 AWG) on the tweeter's 0.75 mm2 flex, pin 1 red +. The socket on the cabinet's lead: receptacle housing 39-01-2020 with female terminals 39-00-0077 (16 AWG) on the cabinet's 1.0 mm2 lead. Mated, 24 long or less.
- 3.2 Bond 3 M2.5 x 4 knurled brass inserts into the seat with epoxy (ø3.6 x 5 holes on ø54; a heat-set insert will not melt into cured resin). Pass the tweeter in from behind through the ø62.4 bore, dome first, seat its faceplate on a 0.5 mm closed-cell foam ring, ID 35 / OD 62, punched (its ID never inside the throat), and screw it to the inserts with M2.5 x 8 button heads (ISO 7380, A2) through its own holes. Measure first: the heads need 0.5 clear of the body (the circle at least the body + 5.7).
- 3.3 Glue the magnets into the insert's back with epoxy, polarity marked, and their partners into the pocket's back wall, opposite poles out. Bond the two pins into the insert with epoxy.
- 3.4 The cabinet's lead is fed down the channel from the bay, through the empty pocket, and caught through the woofer's cut-out (it cannot be pushed up from inside). Its socket stands on the bay's floor, the lead's end 40 mm above the top panel's underside (10 past the floor); seal the channel round it with 20 mm of neutral-cure silicone from the bay before the first fitting. To fit: hold the insert just clear of its pocket, reach in and plug the tweeter's lead into the socket (its 220 mm reaches with a hand in the pocket), then slide the insert home, feeding the spare into the bay as a loop, behind the tweeter's body in the boss's bore and the bay.
- 3.5 Slide the insert in, level, until the pins seat and the magnets pull it home: its face flush with the roof, the seam even. Fit: the pocket +0.2/0 and the insert 0/-0.15 on their widths (0.30 to 0.48 a side; the boss in its bore the same); paint only the insert's face. A print service holds about 0.2 to 0.3 % (0.6 to 0.9 on 159): order the base's width and the boss's ø68.4 as critical dimensions, or sand the unpainted sides, base and boss until it slides home with 0.3 to 0.5 a side; never sand the waveguide, the seat or the bore. Check the width before painting.
- 3.6 Service: pull it out by a ribbon loop glued at the back of the 12 x 1.5 x 30 groove under its front edge, unplug, and the tweeter comes out with it. Between times the loop folds back into its groove: only its end shows, under the insert's front edge.

- 3.7 The pocket (gable-block.step): the insert's outline + 0.3 a side (159.9 wide at its base), from the slope to its back wall at y 113.3; the boss's bore ø69 to y 127; the connector bay ø32 from y 126 to 148 on an axis at z 512; the channel ø14 at x 110, y 137 down to the top panel; R3 or less in the profile's corners only, and sharp where the back wall meets the walls (a flat-ended cutter), as the insert's back edges are; dowel holes ø10 x 20 in the base at (34, 125), (186, 125), (34, 175.5), (186, 175.5). Milled (route A), the back wall is 113.3 and the bay 148 from the front: a cutter with that reach, or print the block (route B).
- 3.8 Fixings, from the centreline and the insert's base (mirrored in the pocket's wall): magnets ±41.4 across, 16.9 up; ±41.5 across, 33.0 up; pins ±46.6 across, 25.9 up. Magnets ø8 x 3 in ø8.2 x 3.3 holes, epoxied, poles marked; pins ø4 x 8 bonded with epoxy in ø4.1 x 4.5 in the insert, a slip fit in ø4.2 x 4.5 in the wall (their own tolerances, not the title block's: the insert's pin holes ø4.1 +0.05/0, the wall's ø4.2 +0.1/0, the magnets' ø8.2 +0.1/0 x 3.3 +0.2/0, each depth from its own face).
- 3.9 Printing the insert: one piece, SLA tough resin or MJF nylon. MJF prints solid; SLA solid, or hollow to 4 mm walls with two ø3 drain holes in the base face between the pins, plugged with resin: none in the waveguide, the seat or the bore, and none within 5 mm of a magnet, pin or insert hole.

Notes: sheet 3

earmilk bookshelf. The notes sheet 3 points to by number: fitting the tweeter and the insert, printing it.

